

Proposal: “Identifying natural images from human brain activity”

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1. Information about the authors of the replication:

- **Jonas Elpelt:**
 - **Institution:** Frankfurt Institute for Advanced Studies
 - **E-mail address:** elpelt@fias.uni-frankfurt.de
 - **fMRI experience:** 2 years fMRI experience.
 - **Big data experience:** Yes.
 - **Research experience:** Familiar with the topic/area of research of this study.
 - **Methodological experience:** Familiar with the analysis methods used.
- **Santiago Galella:**
 - **Institution:** Frankfurt Institute for Advanced Studies
 - **E-mail address:** galella@fias.uni-frankfurt.de
 - **fMRI experience:** 2 years fMRI experience.
 - **Big data experience:** Yes (THINGS, NSD).
 - **Research experience:** Familiar with the topic/area of research of this study.
 - **Methodological experience:** Familiar with the analysis methods used.
- **Abd Alselam Ahmad:**
 - **Institution:** Frankfurt Institute for Advanced Studies
 - **E-mail address:** aahmad@fias.uni-frankfurt.de
 - **fMRI experience:** 3 months of fMRI experience
 - **Big data experience:** Yes (NSD, LAION)
 - **Research experience:** somewhat familiar with the topic/area of research of this study.
 - **Methodological experience:** somewhat familiar with the analysis methods used.

2. Authors of the original study:

Kendrick N. Kay, Thomas Naselaris, Ryan J. Prenger & Jack L. Gallant

3. How would you categorize the scientific subfield of this study:

- Decoding / Reconstruction
- Encoding Models
- Feature Selectivity

4. Dataset / data used in the original study:

- **Dataset:** vim-1 ([Visual Imaging 1](#))
- **Preprocessing:** Released preprocessing.
- **Analysis space:** Native subject space (individual-subject voxel space)
- **Participants:** Two participants. Throughout this proposal, we will refer to them as S1 and S2, respectively.
- **Images:** No fixed semantic subset was used. The study used grayscale natural photographs, with 1,750 images for model training and 120 separate images for testing. The images were not restricted to a specific category.
- **ROIs:** Visual cortex ROIs: V1, V2, and V3.

- **Metadata:** No metadata was used for decoding.

5. Context of the study:

This paper was among the first to demonstrate identification of natural images from human brain activity (fMRI), a task previously limited to simple, low-dimensional stimuli (e.g., gratings) or closed sets of learned categories (e.g., faces, houses). Rather than training a classifier on brain responses to a fixed stimulus set, the authors fit quantitative **receptive-field (encoding) models**, **voxel-wise tuning** for **spatial position**, **orientation** and **spatial frequency** **via a Gabor wavelet pyramid**, from responses to 1,750 natural images, then used these models to predict responses to novel images and identify which one an observer had seen by matching predicted to measured activity. This encoding-model approach was key to generalizing **beyond previously-seen stimuli**, and set the precedent for the model-based decoding / identification framework that much later on has built on (e.g., including deep-network-based encoding models).

6. Main findings of the original study:

1. **Novel-image identification works at high accuracy.** *"We show that these receptive-field models make it possible to identify, from a large set of completely novel natural images, which specific image was seen."* (Fig. 3, correlation matrix, S1, 92% correct at n=120 and Fig. 4a (repeated-trial bars, both subjects, with **noise-ceiling** markers)
2. **Feature tuning (orientation, spatial frequency) affects identification.** *"Identification is not a mere consequence of the retinotopic organization of visual areas; simpler receptive-field models that describe only spatial tuning yield much poorer identification performance".* (Fig. 4c. Gabor model vs. retinotopy-only model).
3. **Performance scales with set size.** (Fig. 4b. performance vs. set size up to 1,000, extrapolated to 10% threshold).
4. **Single-trial identification is possible, though degraded relative to trial-averaged data.** (Fig. 4a. single-trial bars)
5. **Decoding remains stable over time.** No figure panel, reported only in the main text.

7. How were each of these findings generated:

Important

1. **Novel-image identification works at high accuracy** (Fig. 3 and Fig. 4a)
 - **Method:** Voxel-wise Gabor wavelet pyramid (GWP) receptive-field models were fit on the 1,750 model-estimation images (approx **5,000 candidate voxels** in V1-V3; the 500 with the highest predictive power selected, Supp. Fig. 4). For each of the 120 held-out novel images, **the fitted models predicted a voxel activity pattern**. The predicted pattern most correlated (**Pearson's r**) with the measured pattern was selected as the identified image.
 - **Outcome:** 92% (110/120) correct for S1, 72% (86/120) for S2, repeated-trial data, chance=0.8% (1/120).
 - **Statistical result:** **No inferential test on the identification accuracy itself.** The underlying voxel models were validated with a bootstrap one-tailed test of predictive power $r > 0$ (Supp. Methods 6, per-area results in Supp. Table 1). **A noise-ceiling simulation (25 bootstrap-like draws per image from single-trial data, assuming a perfect model) is plotted as markers in Fig. 4a and indicates that the S1/S2 difference reflects noise level rather than model quality.**
2. **Feature tuning (orientation, spatial frequency) affects identification** (Fig 4.c)

- **Method:** A **retinotopy-only (RO)** model, voxel response as a function of only spatially-weighted luminance and contrast in the same estimated receptive-field location, **discarding orientation/spatial-frequency tuning was fit with the same gradient-descent procedure** and **compared to the GWP model**. Voxels were selected **separately** for each model by its own predictive power. **The RO model's** spatial weighting was itself validated: 55% vs. 42% identification at $n=120$ using spatially-weighted vs. standard luminance/contrast metrics, **confirming the RO model was cast in the best possible light before comparison.**
 - **Outcome:** Extrapolated set size at which performance falls to 10%: $10^{9.5}$ images (GWP) vs. $10^{5.1}$ images (RO), four orders of magnitude difference, repeated-trial, averaged across subjects.
 - **Statistical result:** A descriptive comparison of extrapolated performance curves. **The extrapolation procedure itself was validated by checking it against empirically measured performance at a set size of six million images.**
3. **Scaling with set size** (Fig. 4b)
- **Method:** Empirical performance up to $n=1,000$ **was computed as a rank-based probability.** For larger sizes, the distribution of correlations between the pattern and 999 library-image predictions was kernel-smoothed.
 - **Outcome:** Going from 100 to 1,000 images, performance dropped only from 92% to 82% (S1, repeated-trial). Extrapolated 10%-correct set sizes: $10^{11.3}$ (S1 repeated), $10^{7.3}$ (S2 repeated), $10^{5.5}$ (S1 single-trial), $10^{3.5}$ (S2 single-trial).
 - **Statistical result:** The extrapolation's validity rests on the authors confirming it matches empirically measured performance at an independently tested set size of six million images.
4. **Single-trial vs. trial-averaged decoding** (Fig. 4a)
- **Method:** Same identification algorithm applied to voxel patterns averaged over the 13 repeated presentations per image ("repeated-trial") **versus each of the 13 individual single-trial patterns separately** (1,620 single-trial patterns per subject total), both matched against the same 120 GWP-model predictions.
 - **Outcome:** Repeated-trial 92%/72% (S1/S2). Single-trial: 51% ($834/1,620$) / 32% ($516/1,620$) (S1/S2). Chance = 0.8% ($1/120$ repeated; $13.5/1,620$ single-trial)
 - **Statistical result:** Same noise-ceiling simulation as #1 (25 bootstrap-like draws per image), plotted per condition in Fig. 4a.
5. **Decoding stability over time**
- **Method:** The original GWP receptive-field models (estimated once, in the initial sessions) were applied unchanged to data from three additional sessions in S1 only: two sessions approximately 2 months later (together yielding a 120-image set) and one session approximately 14 months later (a 12-image set), all repeated-trial, all images novel.
 - **Outcome:** 82% (99/120) correct at 2 months (chance 0.8%), 100% (12/12) at 14 months (chance 8%).
 - **Statistical result:** Purely descriptive, single subject, single session per interval, no bootstrap/noise-ceiling estimate reported.

8. How to replicate these results:

LAION-fMRI includes repeated presentations of the same natural images to the same subject at a larger scale (5 subjects, 7T, single-trial GLMsingle betas, 6,204 images per subject with individual images repeated 3-12 times across 30 sessions, including 881 images shared and repeated 12 times across all five subjects). **No voxel receptive-field encoding-model pipeline currently exists.**

First, we will build a receptive-field encoding-model pipeline using the Gabor wavelet pyramid, together with a per-voxel fitting and a correlation-based identification procedure, as in the original paper.

1. **Novel-image identification works at high accuracy.** Fit **voxel-wise receptive-field models** on each **subject's non-repeated ("unique") images**, then test identification on held-out repeated trials by predicting voxel activity for the candidate set and selecting the image whose prediction best correlates with the measured pattern.
2. **Feature tuning (orientation, spatial frequency) affects identification.** We will compare the same identification procedure using (a) a spatial-tuning-only model and (b) the full model with orientation/spatial-frequency tuning, to test whether feature tuning beyond retinotopy drives performance. LAION-fMRI's public materials list phase-encoded retinotopy among planned dataset contents, but full retinotopy documentation and processed maps are not yet released. In the meantime, we can use anatomically predicted retinotopy templates (Benson et al., 2014) or deep-learning models (DeepRetinotopy; Ribeiro et al., 2026). For area assignment we can use atlas-based parcellations (Wang et al., 2015; Glasser et al., 2016).
3. **Scaling with set size.** We will use each subject's pool of unique (non-test) images as a distractor library to measure how identification performance degrades as candidate-set size grows, following the same empirical-plus-extrapolation approach as the original study.
4. **Single-trial vs. trial-averaged decoding.** We will compare identification performance using single-trial GLM single betas against activity averaged across a stimulus' repeats, directly analogous to the original single-trial vs. 13-repeat-averaged comparison.

The result "*Decoding remains stable over time*" may not be replicable. In the original study, receptive-field models estimated in the main sessions were applied unchanged to novel images viewed approximately 2 and 14 months later (S1 only). LAION-fMRI has the required structure (30 sessions per subject, models fit on early sessions and tested on later ones, with any later-session images serving as the novel test set), but the released metadata contains no session acquisition dates, so the time gap is currently unknown. We will request session dates from the dataset authors; if the gap is comparable, the analysis is replicable, otherwise it remains unverifiable.

No new metadata has to be generated or approximated for the replication.

9. Statistics:

Kay et al. (2008) report all results per subject (S1, S2) rather than as group-level statistics, so in most cases their approach already satisfies a single-subject design. The one exception is Fig. 4c, where the extrapolated identification thresholds for the Gabor wavelet pyramid (GWP) vs. retinotopy-only (RO) models are averaged across the two subjects into a single pooled number. Separately, several of the paper's headline comparisons (identification accuracy vs. chance; GWP vs. RO; single-trial vs. averaged-trial) are supported only by descriptive percentages in the original, with no formal significance test. For these we add a permutation test within the same single-subject framework.

All permutation tests follow the re:vision procedure: permute along the dimension carrying the effect of interest while holding the rest of the data structure fixed, run 1,000–10,000 iterations per test, and

compute a one-sided p-value. All tests are run and reported separately for each of the 5 subjects. Multiple-comparisons correction is applied within each subject across all permutation p-values generated for that subject in this study.

1. **Novel-image identification works at high accuracy (Fig. 3, Fig. 4a).** No change to the model-level statistic: the original's bootstrap confidence interval on voxel predictive power ($r > 0$, Supp. Method 6) is already per-voxel and per-subject. The headline identification accuracy itself has no formal test in the original (only a descriptive comparison to chance level). We add a permutation test: per subject, shuffle the correspondence between measured voxel-activity patterns and predicted (candidate-image) patterns, recompute % correctly identified under each shuffle to build the null distribution, and compare the observed accuracy against it.
2. **Feature tuning (orientation, spatial frequency) affects identification (Fig 4.c).** The original pools S1 and S2 into a single averaged threshold; we report per-subject results instead. Permutation test per subject: for each candidate image independently, randomly swap which model's (GWP vs. RO) predicted pattern is compared against the measured pattern, recompute the identification-accuracy difference at a fixed set size ($n = 120$) under each swap, and test whether the observed GWP – RO difference exceeds the null.
3. **Scaling with set size (Fig. 4b).** This is an estimation/extrapolation procedure (empirical measurement up to $n = 1,000$, kernel-smoothed extrapolation beyond that, validated against an empirically measured benchmark at $n = 6,000,000$). The original reports no confidence interval on the extrapolated threshold; we add a per-subject bootstrap CI by resampling the distractor library.
4. **Single-trial vs. trial-averaged decoding (Fig. 4a).** We add two permutation tests per subject: (a) accuracy vs. chance for each condition separately (same scheme as Finding 1, run once for single-trial data and once for averaged-trial data); (b) a per-image swap test: on each iteration, draw one random single trial per image, swap its label with that image's averaged-trial pattern, recompute the accuracy difference, and repeat 1,000–10,000 times, to test whether the single-trial accuracy drop exceeds the null.
5. **Decoding stability over time.** If feasible (see Section 8, temporal-stability paragraph): no change to statistical framework needed beyond adding a test, since the original reports this descriptively (single subject, single session per gap). We would apply the same accuracy-vs-chance permutation scheme as in #1, per subject, separately for each available time gap.

10. How to test generalization of these findings:

We carry out full generalization testing (Methods 1, 2 and 3) for the main finding (1, Novel-image identification works at high accuracy) novel-image identification and for the retinotopy-only vs. full-model comparison (2, Feature tuning (orientation, spatial frequency) affects identification), which reuses Finding 1's refitted models at negligible extra cost. The rest of the results are not carried through generalization testing (see below). In all Method 3 analyses, OOD images are excluded from model estimation (models are fit on the standard LAION-fMRI images only).

1. **Novel-image identification works at high accuracy (Fig. 3, Fig. 4a).**
 - o **Method 1 (within-distribution):** Refit the Gabor wavelet pyramid (GWP) encoding model on the LAION-fMRI 80% train split, evaluate identification accuracy on the held-out 20% test split, and compare against the main replication result (fit on unique images, tested on repeated images) to check the result is not specific to that split.

- **Method 2 (OOD clusters):** Refit on 4 of the 5 embedding-space clusters, evaluate identification on the held-out cluster, repeat leave-one-cluster-out across all 5 folds and report the mean. Testing whether performance holds when test images come from visual/semantic content the model never trained on.
 - **Method 3 (OOD images):** Evaluate identification on the cropped and unusual subsets (still photographic but atypical in framing and content, the closest analogue to Kay et al.'s "novel natural images"). The shape subset, non-photographic abstract forms) serves as a more extreme stress test of whether the natural-image-trained model degrades or fails.
2. **Feature tuning (orientation, spatial frequency) affects identification (Fig 4.c)**
 - **Methods 1 and 2.** Refit both the GWP and retinotopy-only (RO) models on the 80% train split (Method 1) and on each leave-one-cluster-out fold (Method 2), and test whether the GWP – RO identification advantage holds on the held-out split / cluster. The GWP refits are shared with Finding 1; only the two-channel RO model is fit additionally.
 - **Method 3,** gabor subset: six localised Gabor patches differing in position, orientation, size and spatial frequency, plus four full-field patterns (left and right spirals, concentric, radial) with matched spatial envelope and contrast. The RO model is invariant to orientation and spatial frequency by construction, so it should perform at or near chance among them (closest to exact for the mirror-image spiral pair), whereas the GWP model can distinguish them only through voxel orientation tuning. The prediction is GWP above chance and RO at or near chance on the confusion among the four full-field patterns (chance 25% within that subset, 10% for the full 10-way identification), with both models able to identify the six positioned patches.
 3. **Scaling with set size.** This is an estimation/extrapolation procedure over a large distractor library rather than a content-driven effect; Methods 1–3 would test the stability of the extrapolation across image subsets rather than generalization in the intended sense, at high computational cost.
 4. **Single-trial vs. trial-averaged decoding.** This reflects fMRI measurement noise rather than content-dependent encoding and is not expected to depend on which images are held out; a Method 1 check is optional if time allows.
 5. **Decoding stability over time.** This concerns generalization across time, not image content; Methods 1–3 do not naturally apply, consistent with the guide's note that they may not fit findings defined over a restricted stimulus dimension.